

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech (CL) Examination

March - April 2018

CL741 Advanced Structural Analysis

Date: 29.03.2018, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Uses of **MS Excel, Scilab and MATLAB** are allowed.

SECTION – I

Q – 1 (a) Enlist the characteristics of flexibility matrix. [03]

Q – 1 (b) Explain following terms with necessary diagrams. [04]

(i) Unstable structures

(ii) Degree of freedom

(iii) Member end actions

(iv) Statically indeterminate structures

Q – 2 Analyse the continuous beam shown in **Figure 1** using stiffness method member approach. Consider I as constant throughout the span. Also draw Shear Force and Bending Moment diagrams. [14]

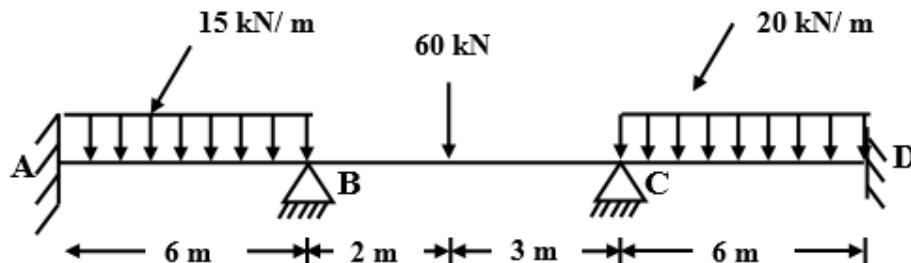


Figure: 1

OR

Q – 2 (a) Develop member stiffness matrix for a space truss member in structural axes with necessary diagrams. [07]

Q – 2 (b) Explain rotation of axes for two-dimensional member with neat diagram. [07]

Q – 3 (a) Explain unsymmetry by giving one example of beam and plane frame. [04]

Q – 3 (b) Analyse the beam shown in **Figure 2** by flexibility method system approach. Draw shear force diagram. [10]

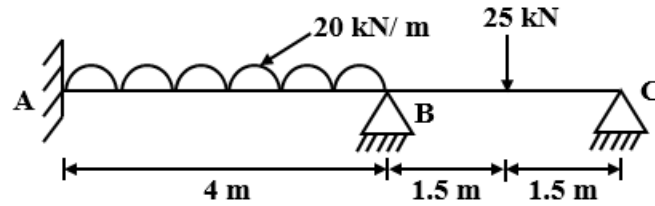


Figure: 2

OR

- Q - 3** Analyse the plane truss shown in **Figure 3** using direct stiffness method. Consider **[14]**
 $E = 200 \text{ GPa}$ and $A = 1000 \text{ mm}^2$ for all members. Also draw axial force diagram.

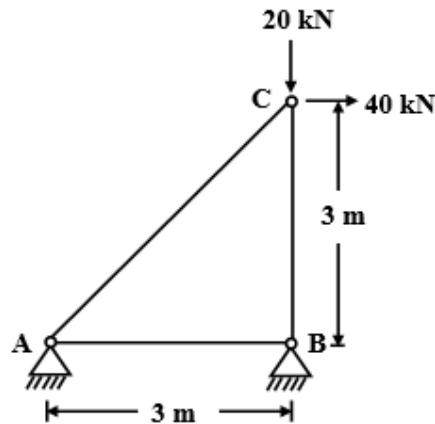


Figure: 3

SECTION – II

- Q – 4** Write member stiffness matrix for space frame member shown in **Figure 4**. Consider **[07]**
 the displacements shown in figure only and structural axes as reference axes. Take L , GJ , EI and EA as constant.

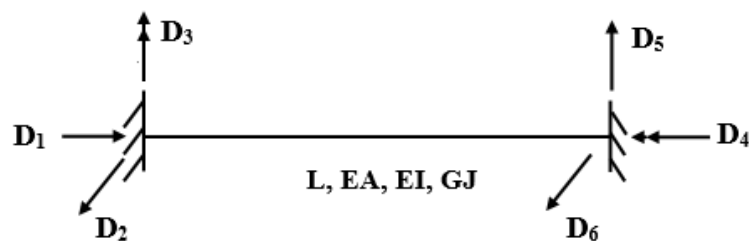


Figure: 4

- Q – 5** Calculate support reactions for the given plane frame shown in **Figure 5** by stiffness **[14]**
 method member approach. Take $E = 200 \text{ GPa}$, $b = 230 \text{ mm}$ and $d = 450 \text{ mm}$. Also
 draw shear force diagram.

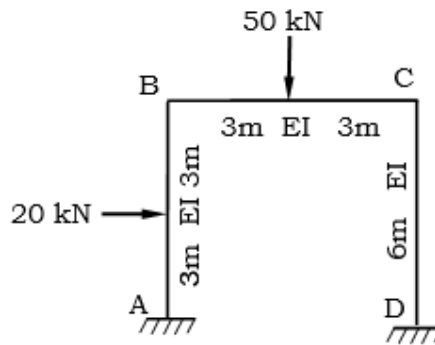


Figure: 5

OR

- Q – 5** Find unknown reactions the portal frame shown in **Figure 6** by direct stiffness method. Take $E = 200 \text{ GPa}$, $b = 230 \text{ mm}$ and $d = 450 \text{ mm}$. Also draw shear force diagram.

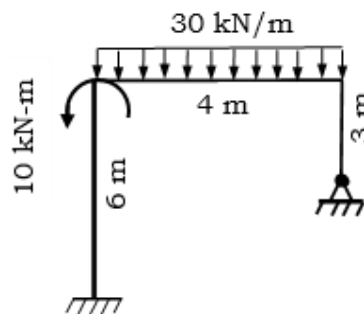


Figure: 6

- Q – 6** Analyse the grid shown in **Figure 7** using stiffness method member approach. [14] Consider I as constant for all members. Assume $EI = GJ$. Note that the axes shown in figure indicate directions only. Draw shear force diagram and bending moment diagram.

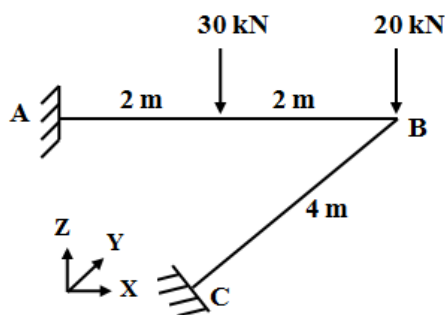


Figure: 7

OR

- Q – 6** Analyse the grid shown in **Figure 8** using stiffness method member approach. [14]
Consider I as constant for all members. Assume $EI = GJ$. Note that the axes shown in figure indicate directions only. Draw shear force diagram.

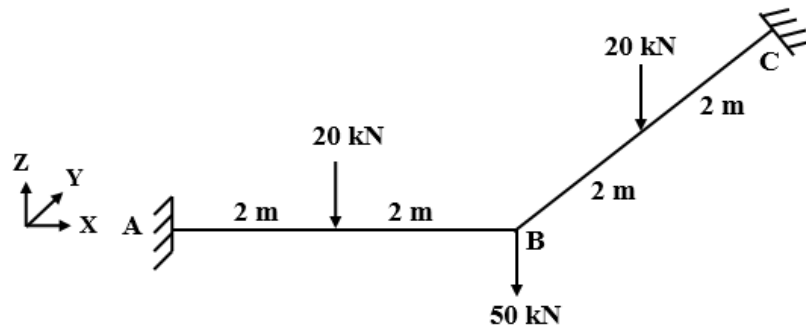


Figure: 8
